

NIS2

COMPLIANCE FRAMEWORK



Terraform

AWS

NIS2

THE PROBLEM



Why NIS2 Compliance matters right now ?

29,000+

New organizations
regulated in Europe

€10M

Maximum fine for
non-compliance

24h

Time limit to report
a security incident

NIS2 (EU 2022/2555) entered into force October 2024. Most organizations still lack automated compliance infrastructure — creating urgent demand for cloud security engineers.

THE SOLUTION



4 architectures on our environment cloud

01

Identity & Access

MFA enforcement
Password policy
Least privilege

Art.21 §2(j)

02

Audit Logging

CloudTrail
KMS encryption
S3 × 365 days

Art.21 §2(h)

03

Threat Detection

GuardDuty
Security Hub
8 Config Rules

Art.21 §2(b)

04

Incident Alerting

CloudWatch
SNS alerts
4 scenarios

Art.23 §1

NIS2 COMPLIANCE MAPPING



NIS2 Article	Requirement	AWS Service	Status
Art. 21 §2(j)	Access control & MFA	IAM + MFA Policy	Done
Art. 21 §2(h)	Cryptography	KMS + S3 SSE	Done
Art. 21 §2(a)	Risk analysis & audit	CloudTrail + Config	Done
Art. 21 §2(b)	Incident handling	GuardDuty + SNS	Done
Art. 21 §2(f)	Continuous assessment	Security Hub + 8 Rules	Done
Art. 23 §1	Incident notification 24h	CloudWatch + SNS	Done

TECH STACK



Terraform

IaC v1.15.7



AWS IAM

Access Control



CloudTrail

Audit Trail



GuardDuty

Threat Detection



Security Hub

Posture Mgmt



AWS Config

8 Compliance Rules



KMS

Encryption



CloudWatch

Monitoring

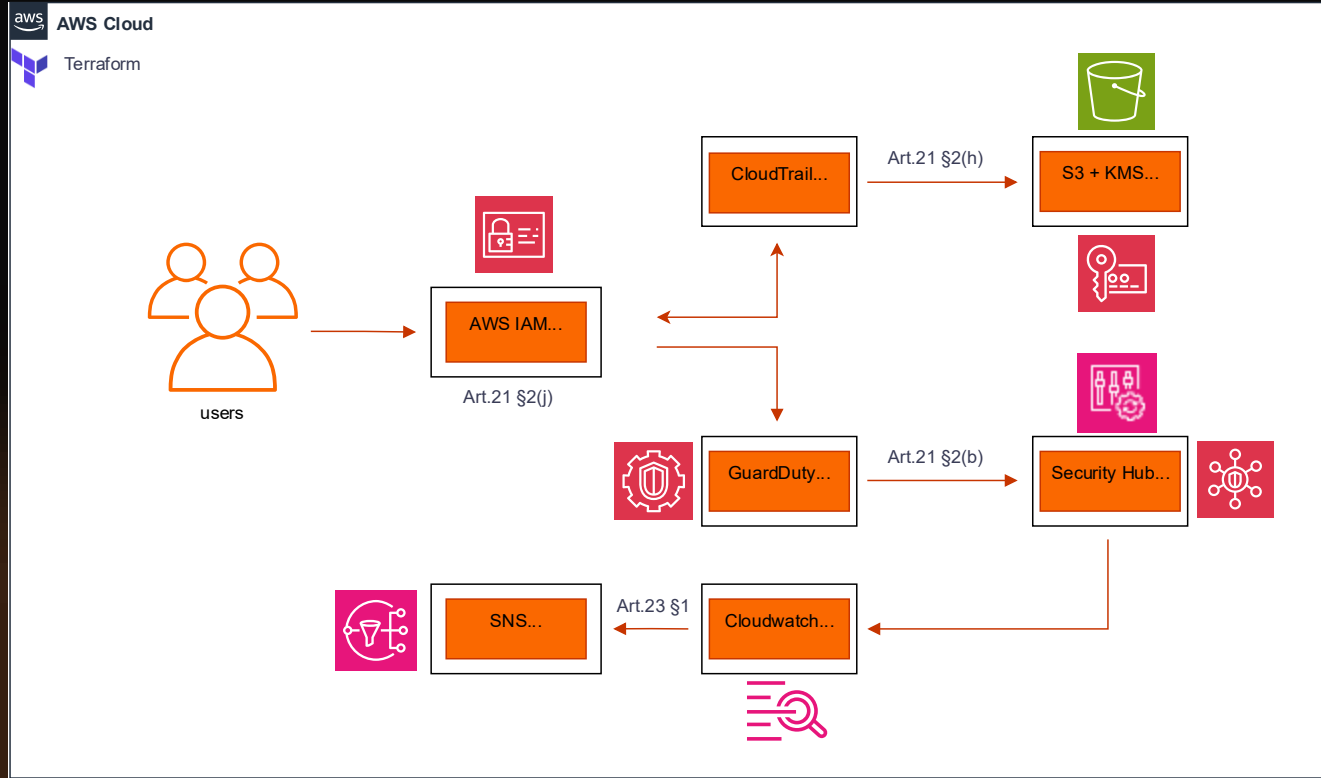


SNS

Alerting



ARCHITECTURE ON AWS



VIEW OF OUR CONSOLE AWS



```
PS C:\... \Desktop\nis2-compliance-framework> tree
Structure du dossier
Le numéro de série du volume est 00000052 9082:5257
C:\.
├── terraform
│   ├── modules
│   │   └── providers
│   └── providers
├── compliance-mapping
├── docs
├── modules
│   ├── alerting
│   │   ├── terraform
│   │   │   └── providers
│   │   └── providers
│   ├── modules
│   │   └── alerting
│   ├── detection
│   ├── iam
│   └── logging
```

Tree of our Project

GuardDuty > Résultats

Résultats (2) Infos

Créer une règle de suppression

Actions

Saved filters

Apply saved filters

Filtrer les résultats

Statut

Actuels

Type de menace

Tous les résultats

Titre

The API GetInvitationsCount was

The API GetAccountEmailAddress

Action type	AWS_API_CALL
API	GetFindingsStatistics
Service name	guardduty.amazonaws.com
First seen	26-06-2026 04:25:29 (il y a 36 minutes)
Last seen	26-06-2026 04:52:11 (il y a 10 minutes)
Acteur	
Caller type	Remote IP
IP address V4	192.168.1.1
Location	
City	Paris
Country	France
Lat	48.8566
Lon	-2.3522
Organization	
Asn	16509
Asn org	OVH
Isip	192.168.1.1
Org	OVH SAS
Informations supplémentaires	
Archived	false

View on GuardDuty

VIEW OF OUR CONSOLE AWS



CloudWatch

Favoris et récents

Ingestion

Tableaux de bord

Alarmes 0 4 0

Opérations liées à l'IA

GenAI Observability

Vigie applicative (APM)

Surveillance de l'infrastructure

Journaux

Métriques

Surveillance du réseau

Configuration

Ateliers sur l'observabilité

Alarmes Règles de désactivation - nouveau

Alarmes (4)

Filtrer les alarmes

État = OK Effacer les filtres

	Nom	État	Actions
☐	nis2-framework-iam-policy-changed	OK	Actions activées
☐	nis2-framework-root-login-detected	OK	Actions activées
☐	nis2-framework-cloudtrail-tampered	OK	Actions activées
☐	nis2-framework-s3-bucket-policy-changed	OK	Actions activées

View on CloudWatch

AWS

CloudTrail

Journal de suivi

arn:aws:cloudtrail:eu-west-1:123456789012:trail/nis2-framework-trail

CloudTrail

Tableau de bord

Event coverage

Historique des événements

Insights

Journal de suivi

Like

Tableaux de bord

Requête

Stockages de données

Intégrations

Paramètres

nis2-framework-trail

Détails généraux

Journalisation de suivi

Nom du journal de suivi

Journal de suivi multirégion

Appliquer un journal de suivi à mon organisation

Emplacement du journal de suivi

Dernier fichier journal livré

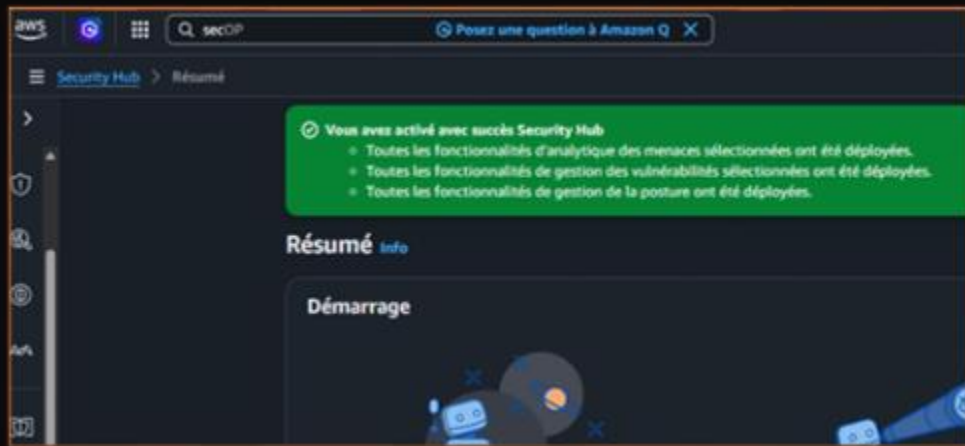
Chiffrement SSE-KMS du fichier journal

Clé AWS KMS

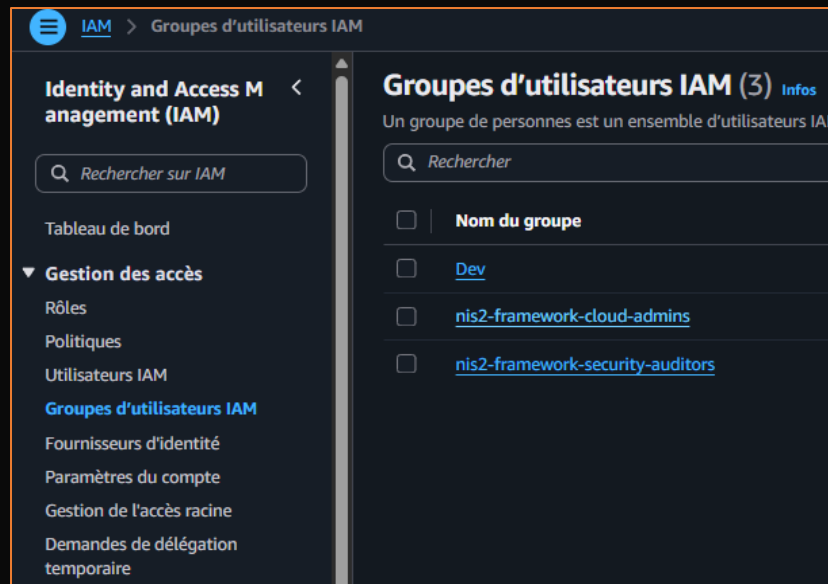
Alias de clé AWS KMS

View on CloudTrail

VIEW OF OUR CONSOLE AWS



View on Security Hub



View on IAM

CONCLUSION



This project demonstrates a production-ready NIS2 compliance infrastructure built entirely with **Terraform** and **AWS**. All resources are version-controlled, documented, and mapped to specific **NIS2 directive articles**.

AREAS FOR IMPROVEMENT



Art.21 §2(c) — Business Continuity

Remote backend Terraform

Dedicated VPC

CloudWatch alarms connected on CloudWatch Log Group CloudTrail

SOURCES



<https://www.enisa.europa.eu/publications/nis2-technical-implementation-guidance>
<https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A32022L2555>
<https://registry.terraform.io/providers/hashicorp/aws/latest/docs>
<https://docs.aws.amazon.com>



THANK YOU



[Linkedin](#)



Stratecloud.com



github.com/Naaka01